Population Genetics

BME 105

Terms used to describe populations

Population– a group of interbreeding individuals of a single species living in the same time and place

Gene pool– the total of all alleles carried in all members of a population

Sample – a number of individuals used to make inferences about the entire population

Phenotype and genotype frequency

Phenotype frequency – proportion of individuals in a population that have a particular phenotype

Genotype frequency – proportion of individuals in a population that carry a particular genotype

Example: A gene with two alleles (A and a) in a population of 20 individuals

12 are AA 4 are Aa 4 are aa

Genotype frequencies: AA = 12/20 = 0.6

Aa = 4/20 = 0.2

aa = 4/20 = 0.2

Calculating allele frequencies

Allele frequency – proportion of gene copies in a population that are of a given allele type

Allele frequencies: in 20 people, there are a total of 40 alleles

12 AA individuals 24 A alleles

4 Aa individuals 4 A alleles and 4 a alleles

4 aa individuals 8 a alleles

Frequency of A alleles = (24 + 4)/40 = 0.7

Frequency of a alleles = (8 + 4)/40 = 0.3

General formula for calculating allele frequency

Frequency of A = Frequency of AA + (1/2) frequency of Aa

Frequency of a = frequency of aa + (1/2) frequency of Aa

From genotype frequencies to allele frequencies

Genotype	*AA*	*Aa*	*aa*	Total
Number of individuals	12	4	4	20
Genotype frequency	0.6	0.2	0.2	1.0

Allele	*A*	*a*	Total
Number of chromosomes	28	12	40
Allele frequency	0.7	0.3	1.0

The Hardy-Weinberg equilibrium (HWE) correlates allele and genotype frequencies

Developed independently in 1908 by G.H. Hardy and W. Weinberg

Five simplifying assumptions:

The population has an infinite number of individuals

Individuals mate at random

No new mutations appear

No migration into or out of the population

Genotypes have no effect on ability to survive and transmit alleles to the next generation

Predicting genotype frequencies from allele frequencies with HWE

In a large population of randomly breeding individuals with no new mutations, no migration, and no differences in fitness based on genotype:

$$p^2 + 2pq + q^2 = 1$$

p = frequency (A)

q = frequency (a)

Allele frequencies

p = freq. (A) = 0.7

q = freq. (a) = 0.3

		Eggs	
		p	q
		0.7	0.3
Sperm	p 0.7	AA 49%	Aa 21%
	q 0.3	Aa 21%	aa 9%

Homozygote $AA = \mathbf{p^2} = 0.49$

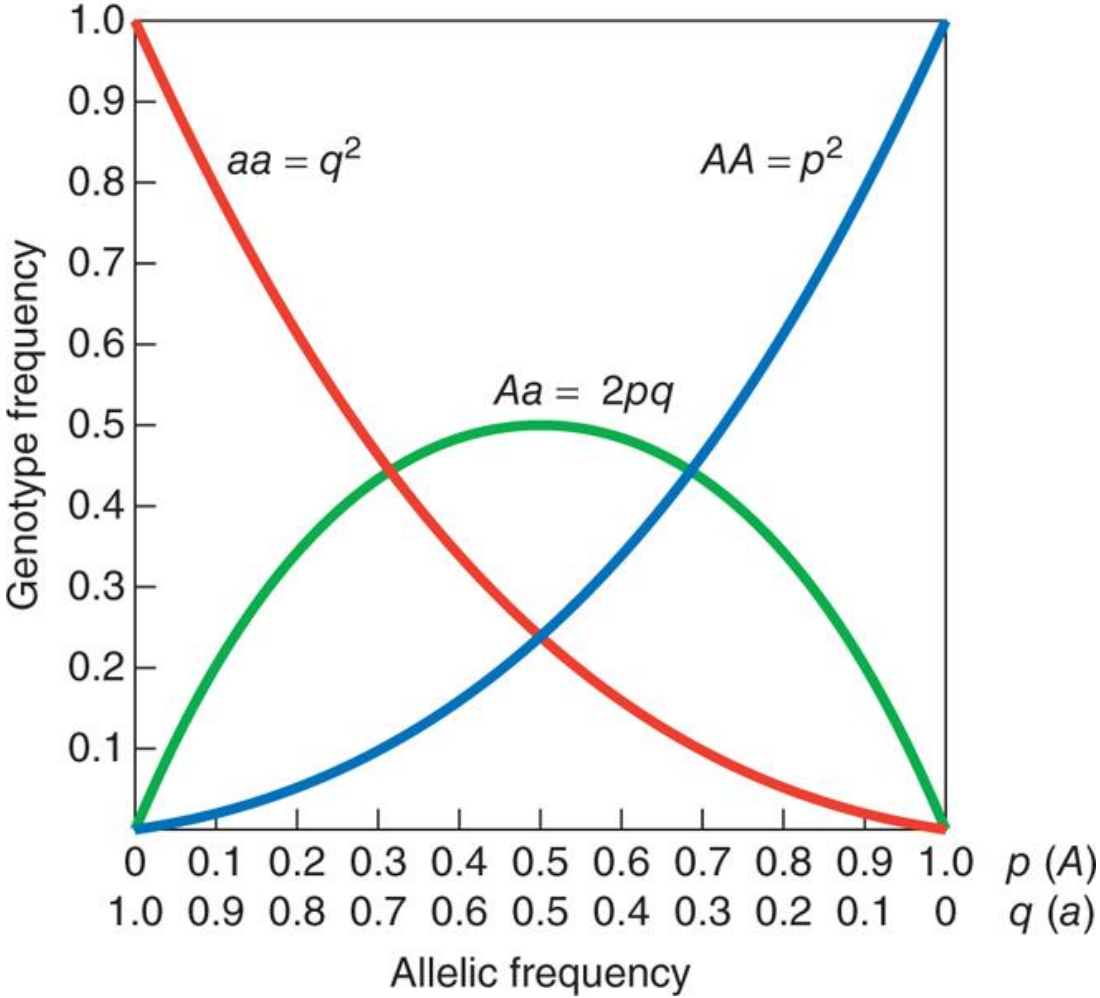
Heterozygote $Aa = \mathbf{2(pq)} = 0.42$

Homozygote $aa = \mathbf{q^2} = 0.09$

1.00

For any set of allele frquenices, only one set of genotype frequencies results in HWE

Assumptions of HWE have to hold



How do you get changes in allele frequencies

HWE is accurate in predicting allele frequency in the short term rather than the long term

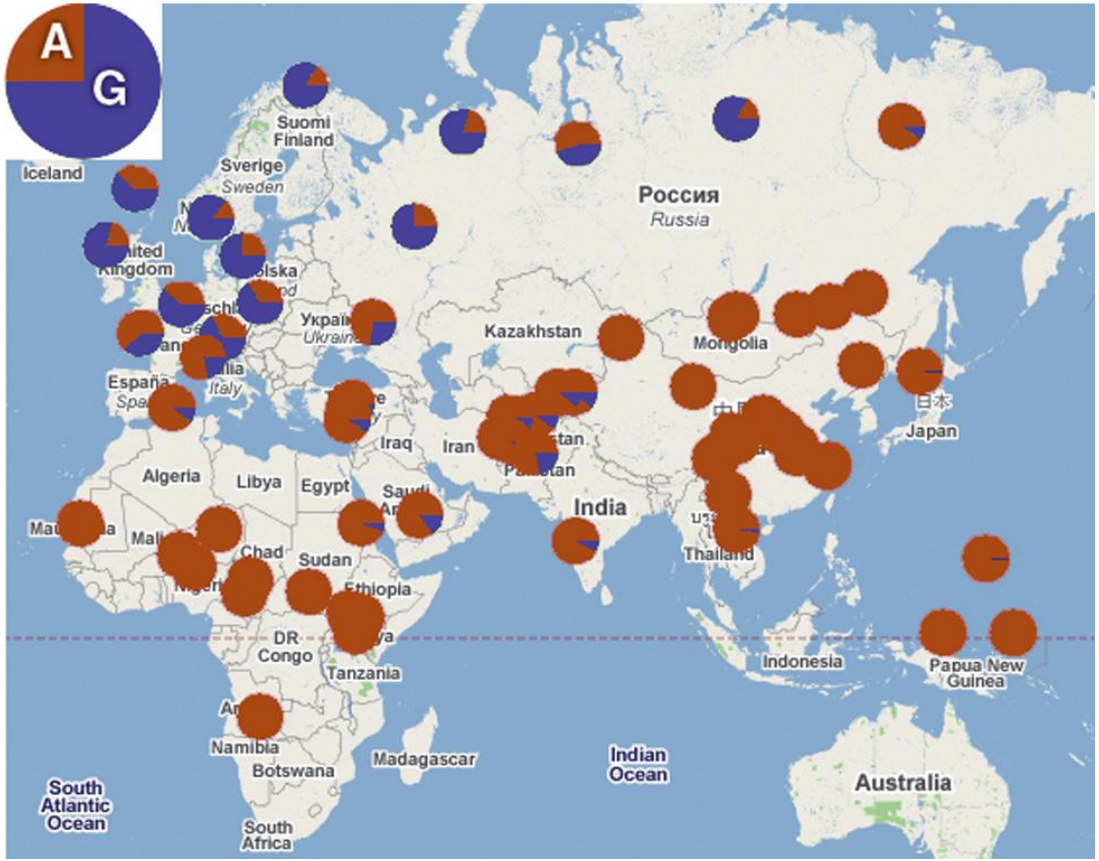
Other factors will cause changes in allele frequency:

- Finite population sizes
- Natural selection

Frequencies of the A and G alleles of the OCA2 eye color gene vary in different populations

p = A allele = brown

q = G allele = blue



Monte Carlo simulations to model long-term changes in allele frequencies with finite population size

Monte Carlo simulations use a computer program to model possible outcomes of randomly chosen matings over a designated number of generations

Starting population has a defined number of individuals that are homozygous and heterozygous

Mating pairs are chosen through a random-number generating program

Genotypes of offspring at each generation are based on probabilities

Monte Carlo simulations to model long-term changes in allele frequencies with finite population size

At each generation in the simulation:

Total offspring number and parental population size are equal

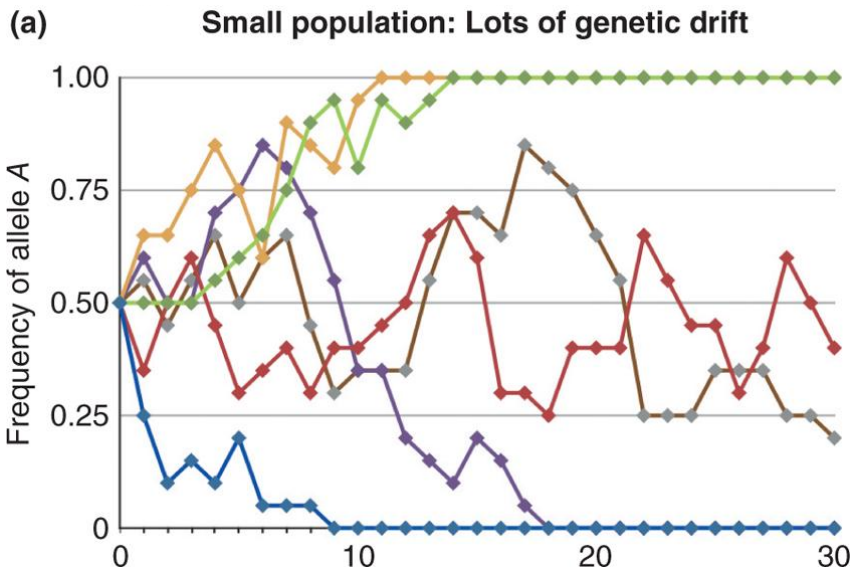
Parental generation is discarded and offspring serve as parents of next generation

Multiple, independent simulations are performed

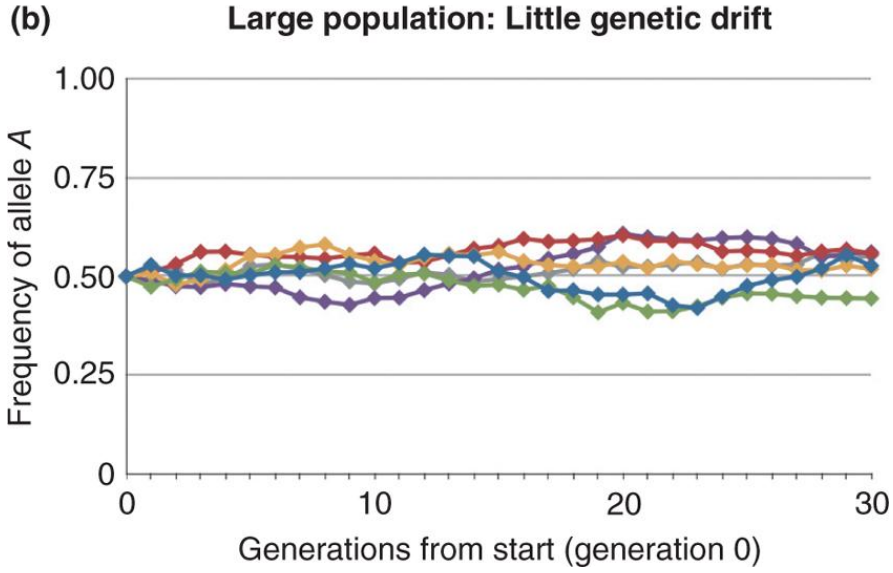
Each simulation represents a possible pathway of **genetic drift**

Monte Carlo modeling of population drift in populations of different sizes

Initial population has 10 individuals



Initial population has 500 individuals



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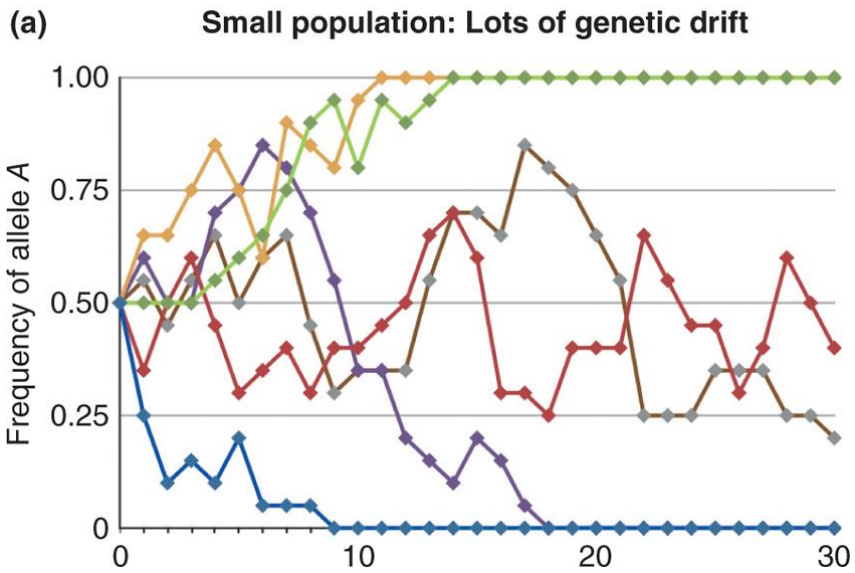

Fixation - when only one allele in a population has survived

Natural selection effects allele frequency

Natural selection in pocket mice

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Modification to HW to account for natural selection

In populations undergoing selection, each genotype has a relative fitness

e.g. Population with two allele (*R* and *r*)

Relative fitness (W) of each genotype (*RR*, *Rr*, and *rr*):

W_{RR} W_{Rr} W_{rr}

Relative frequencies of each genotype at adulthood:

$p^2 W_{RR}$ $2pqW_{Rr}$ $q^2 W_{rr}$

Fitness-modified HWE

Average fitness of the population:

$$\overline{W} = p^2 W_{RR} + 2pqW_{Rr} + q^2 W_{rr}$$

Changes in allele frequencies caused by selection

Parental Gametes

Allele	R	r
Frequency	p	q

First Generation Offspring

Genotypes	RR	Rr	rr
Zygote frequency	p^2	$2pq$	q^2
Relative fitness	W_{RR}	W_{Rr}	W_{rr}
Relative frequency after selection	p^2W_{RR}	$2pqW_{Rr}$	q^2W_{rr}
Normalized adult frequency	$\frac{p^2W_{RR}}{\overline{W}}$	$\frac{2pqW_{Rr}}{\overline{W}}$	$\frac{q^2W_{rr}}{\overline{W}}$

Gametes from First Generation

Allele	R	r
Frequency	$p' = \frac{(p^2W_{RR} + pqW_{Rr})}{\overline{W}}$	$q' = \frac{(q^2W_{rr} + pqW_{Rr})}{\overline{W}}$

Normalization factor $\overline{W} = p^2W_{RR} + 2pqW_{Rr} + q^2W_{rr}$

Calculating the changes in allele frequencies due to selection

p' and q' represent allele frequencies after one generation of selection

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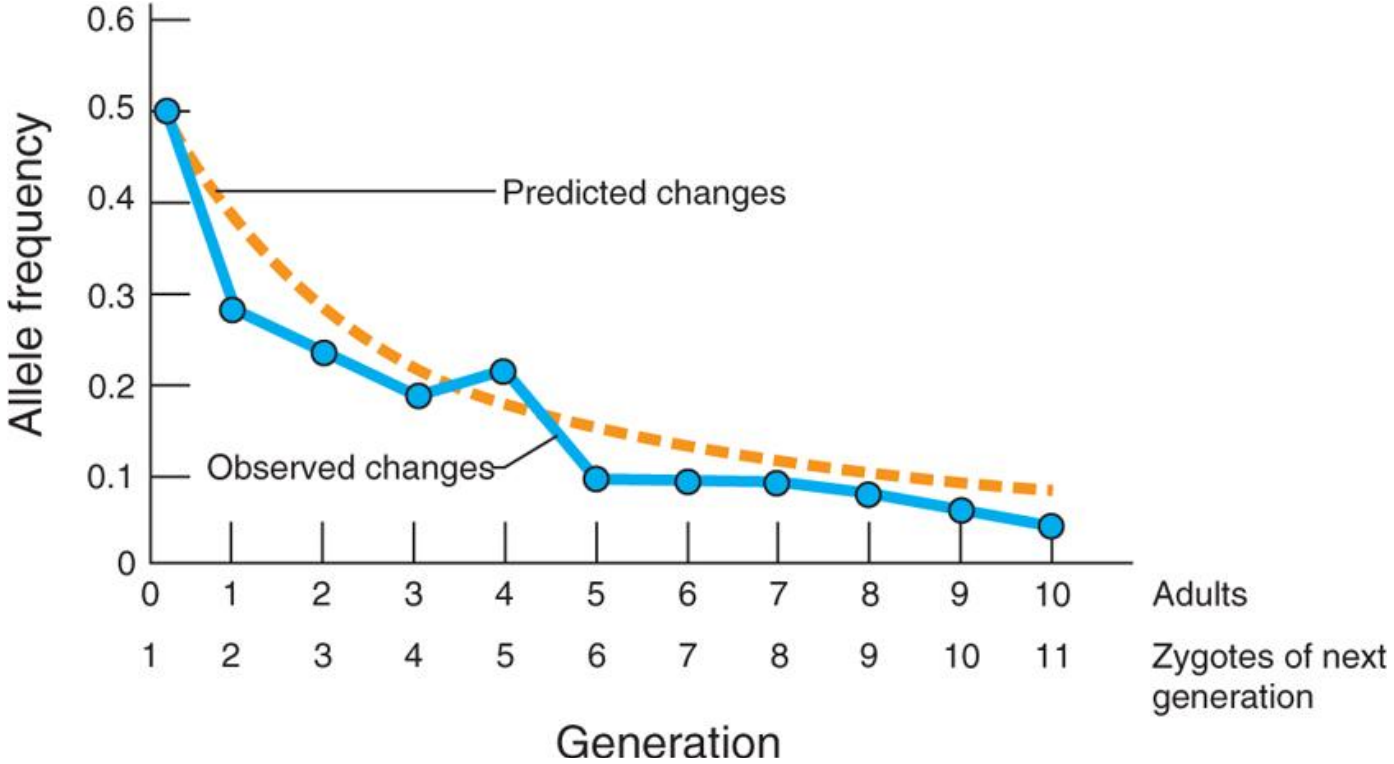
$$q' = \frac{q^2 W_{aa} + \frac{1}{2}(2pqW_{Aa})}{\overline{W}} = \frac{q(qW_{aa} + pW_{Aa})}{\overline{W}} \qquad (20.5)$$

$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\overline{W}} \qquad (20.6)$$

Rate of decrease depends on the allele frequencies

As q approaches 0, rate of decrease gets slower

Decrease in the frequency of a recessive lethal allele over time